TOWER DOMINION / STRATEGY RESEARCH

MISSION CONTROL
CHAIN ANALYSIS

A simulation-based comparison of Blueprint acquisition, faction and commander advantages, and Amplifier effects.

10,000 simulated runs | Waves 1–103

MODELED ACQUISITION — NOT COMBAT SURVIVAL

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Scope: Mission Control acquisition, not survival probability. All scenarios use the same 1,000-run simulation framework.

SECTION 1 · INTRODUCTION

1.1. What is a Mission Control chain?

A Mission Control chain is a strategy built around acquiring multiple Mission Control buildings and using their upgrades to increase the likelihood of obtaining additional Mission Controls.

The strategy relies on two complementary effects.

Mission Control's 1A upgrade increases the Blueprint rarity weights of Tier 2 and Tier 3 buildings. This improves the weight of Mission Control itself, although it also increases the weights of competing Blueprints in those tiers.

The 3A upgrade provides two additional Blueprint drafts, each offering five choices. When Mission Control appears among those choices, the player can acquire another copy, construct it, and eventually use its own upgrades to continue the process.

Once the player can repeatedly afford these upgrades, each additional Mission Control creates opportunities to acquire more. This recursive interaction is what we refer to as the Mission Control chain.

1.2. Why pursue multiple Mission Controls?

Mission Controls provide a way to increase the number of available buildings beyond what ordinary reinforcement rewards would typically provide.

A successful chain can generate additional Blueprint opportunities, allowing the player to expand their defenses and acquire buildings that might otherwise be difficult to obtain.

However, the strategy requires an initial investment. Mission Controls must first be acquired, constructed, and upgraded before their strongest acquisition effects become available.

Consequently, the timing of the first few Mission Controls is particularly important. An early acquisition can provide more opportunities for growth, while a late acquisition leaves fewer waves for the chain to develop.

1.3. The importance of Wave 63

Around Waves 63–64, Tower Dominion introduces a significant increase in enemy wave size. This transition can make it considerably more difficult to continue a run without an established defensive advantage.

For a Mission Control strategy, this creates a practical deadline.

The objective is not simply to accumulate as many Mission Controls as possible by the end of a run. It is to establish a productive chain early enough that the additional buildings can contribute to the player's defenses before this difficulty increase.

For that reason, this analysis treats Wave 63 as its principal evaluation point.

Wave 60 serves as an earlier checkpoint, while Waves 83 and 103 are used to examine how the chain continues developing over a longer run.

1.4. Objectives of the analysis

This analysis investigates how Mission Control acquisition develops over time and how different scenarios influence that process.

The principal questions are:

- How quickly can a Mission Control chain become established?

- How does access to 3A upgrades at Wave 40 change the acquisition rate?

- How likely is a player following the modeled strategy to acquire 5, 10, or 20 Mission Controls by Wave 63?

- How do Lions, Eleonor, Tander, and Magel compare with the baseline scenario?

- How much do Amplifiers contribute to the growth of the chain?

- How does Mission Control acquisition continue through Waves 83 and 103?

The analysis focuses on the acquisition process rather than directly measuring combat effectiveness or the probability of surviving individual waves.

SECTION 2 · HOW THE SIMULATION WORKS

2.1. Overview of the simulation model

The analysis uses a Monte Carlo simulation to reproduce the acquisition of Mission Controls over a series of waves.

Rather than assuming a fixed number of Mission Controls will be obtained, the simulation generates reinforcement rewards and Blueprint drafts using weighted probabilities derived from the game's data and the mechanics incorporated into the model.

Each simulated run begins at Wave 1 and progresses through Wave 103.

During each wave, the simulation accounts for the available reward opportunities, the current Blueprint weights, acquired Doctrines, and the effects of previously constructed Mission Controls.

As the run progresses, the available rewards and their relative probabilities can change.

To compare strategies, each scenario is simulated 1,000 times. The resulting data allows us to examine average acquisition rates and the frequency with which different Mission Control thresholds are reached.

The model represents a player actively pursuing Mission Controls. It does not attempt to simulate a player selecting reinforcement rewards randomly.

2.2. Blueprint acquisition and reinforcement

Mission Control is a Tier 3 Blueprint that can appear in ordinary reinforcement rewards and certain additional Blueprint drafts.

The simulation uses a pool of 32 ordinary reinforcement-eligible Blueprints, distributed across three tiers:

Tier 1: 12

Tier 2: 10

Tier 3: 10

Blueprints have individual rarity weights that change as the run progresses.

Importantly, the probability of obtaining Mission Control is determined by its weight relative to the combined weights of other eligible rewards. Increasing Mission Control's weight does not guarantee that it will appear.

Ordinary reinforcement presents three cards, each containing two rewards. The player selects one card and receives both rewards.

The simulation uses a selection policy that prioritizes cards containing Mission Control. When Mission Control is unavailable, the model can favor rewards that support future acquisition, including an initial Amplifier.

This creates a consistent strategy across simulated runs while preserving the randomness of the available rewards.

2.3. Mission Control upgrades: 1A and 3A

The simulation incorporates two upgrades that are central to the strategy.

1A — Blueprint rarity increase

Each constructed Mission Control contributes an additional two points of raw rarity weight to eligible Tier 2 and Tier 3 Blueprints.

These contributions accumulate as additional Mission Controls are built.

Because the bonus applies to every eligible Blueprint in those tiers, the effect is not equivalent to directly increasing Mission Control's selection probability by a fixed percentage.

3A — Additional Blueprint drafts

Each Mission Control upgraded along the 3A path provides two Blueprint drafts, with five Blueprint choices in each draft.

When Mission Control appears among the available choices, the simulation selects it.

This can produce additional Mission Controls, which can then be upgraded to generate further drafts.

For the standard scenarios, the simulation assumes that the player can afford to upgrade all constructed Mission Controls to 3A beginning at Wave 40.

This is an economic simplification intended to isolate the acquisition process. It does not mean that every actual run will have sufficient resources to upgrade every Mission Control at that point.

2.4. Amplifiers and Blueprint probability

Amplifiers can increase the rarity weight of a targeted building's Blueprint.

When assigned to Mission Control, each Amplifier provides a +100% modifier to its Blueprint weight.

These modifiers stack additively:

0 Amplifiers: 1x

1 Amplifier: 2x

2 Amplifiers: 3x

The model includes two independent Amplifier-granting Doctrines.

An Amplifier can be acquired before the first Mission Control appears. In that situation, it can remain in inventory until a Mission Control is available to receive its effect.

The standard simulation uses naturally acquired Amplifiers rather than assuming that the player begins with one.

Although the model allows both Amplifiers to stack, its current reward-selection policy does not systematically prioritize obtaining the second Amplifier after the first has been acquired.

2.5. Acquisition priorities and building assumptions

The simulation follows a consistent acquisition strategy intended to support Mission Control growth.

When a reinforcement reward offers Mission Control, the model prioritizes acquiring it. Mission Controls are then constructed immediately, activating their modeled upgrade effects.

Before Wave 40, constructed Mission Controls contribute their 1A rarity-weight bonuses.

Beginning at Wave 40, the model also applies their 3A effects and resolves the resulting Blueprint drafts, including any additional drafts generated through newly acquired Mission Controls.

Amplifiers are likewise constructed and assigned to Mission Control when possible.

This approach removes several practical constraints that may affect actual gameplay, particularly construction timing, building placement, and resource availability.

It is therefore best understood as a model of acquisition potential under the specified strategy, rather than a complete simulation of a playable run.

2.6. Simulation parameters and limitations

Starting wave: 1

Final wave: 103

Simulations per scenario: 1,000

3A upgrade availability: Wave 40

Main evaluation point: Wave 63

Amplifier acquisition: Natural, or Tier I Amplifier Doctrine owned from Wave 1

Maximum modeled Amplifier effect: Two stacking Amplifiers

Mission Control construction: Immediate

Scenarios: Baseline, Lions, Eleonor, Tander, Magel

The simulation does not currently account for every mechanic that could influence Mission Control acquisition.

In particular, the Pit, Reactive, and Calculator effects remain outside the model. The simulation also does not reproduce complete construction costs, placement constraints, or the decision to hold additional Mission Controls in inventory.

Some reward-generation details and faction- or commander-specific effects remain provisional.

These limitations are important when interpreting the results. The simulation can estimate how frequently a particular Mission Control count is reached under its assumptions, but it cannot directly determine whether a player will survive Wave 63.

SECTION 3 · ESTABLISHING A MISSION CONTROL CHAIN

3.1. Acquiring the first Mission Control

The first Mission Control is an important milestone because it allows the player to begin accumulating the Blueprint rarity bonuses provided by its 1A upgrade.

Before acquiring one, the player depends primarily on ordinary reinforcement rewards and any additional Blueprint opportunities available through the selected commander. Mission Control is a Tier 3 Blueprint; its rarity weight increases as waves progress.

An Amplifier-granting Doctrine acquired before the first Mission Control can also help. Once Mission Control is available, the Amplifier can be assigned to it, increasing its Blueprint weight.

The timing of these early acquisitions introduces considerable variation. Some runs establish the foundation of a chain early, while others reach Wave 40 with few or no Mission Controls.

3.2. Growth before Wave 40

Before Wave 40, the model permits Mission Controls to contribute their 1A rarity-weight bonus, but 3A Blueprint drafts are not yet available. Growth primarily depends on reinforcement rewards, faction- or commander-specific opportunities, and acquired Doctrines.

Each constructed Mission Control adds two raw rarity-weight points to eligible Tier 2 and Tier 3 Blueprints. Because the bonus also affects competing Blueprints, it is not a fixed percentage increase to the probability of finding Mission Control.

Average Mc Count At Wave 40 — Natural Acquisition (1,000 Runs)

Baseline: 2.206

Lions: 4.811

Eleonor: 3.480

Tander: 3.262

Magel: 2.822

These are end-of-Wave-40 counts, including the 3A drafts activated on Wave 40. They should not be described as counts immediately before the Wave-40 transition.

Lions leads at this checkpoint, followed by Eleonor, Tander, Magel, and Baseline. The faction and commander bonuses influence different acquisition channels: Lions grants additional ordinary Blueprints; Eleonor provides periodic Blueprint drafts; Tander presents an additional reinforcement card; and Magel provides a choose-one-of-three Doctrine draft every other wave.

3.3. The impact of 3A upgrades beginning at Wave 40

Wave 40 is the principal transition in the modeled strategy. At this point, the simulation assumes all constructed Mission Controls can be upgraded to 3A, each generating two additional choose-one-of-five Blueprint drafts.

If a draft offers Mission Control, the model selects and constructs it immediately, then applies its eligible upgrades. That new Mission Control can generate further drafts, creating a recursive acquisition sequence within the same wave.

This assumption isolates acquisition potential: it does not account for the actual resources, construction space, or time required to build and upgrade every copy.

3.4. Recursive Blueprint acquisition

Consider a player with three constructed Mission Controls entering Wave 40. Upgrading all three to 3A produces six Blueprint drafts. If a draft offers another Mission Control, that new copy can produce two more drafts; subsequent successes may extend the sequence.

Each branch can also end without finding Mission Control. Growth therefore remains probabilistic, depending on the number of drafts generated and Mission Control's weight relative to competing Blueprints.

Average Mc Counts — Natural Acquisition (1,000 Runs)

Scenario	W40	W50	W60	W63
Baseline	2.206	4.238	7.147	8.310
Lions	4.811	9.909	16.187	18.507
Eleonor	3.480	6.764	11.338	12.643
Tander	3.262	6.020	10.007	11.410
Magel	2.822	5.739	10.290	12.093

Lions retains the strongest average acquisition rate. Eleonor remains ahead of both Tander and Magel through Wave 63. Magel begins behind Tander at Wave 40, but moves ahead by Wave 60. Its extra Doctrine drafts offer more chances to acquire the Amplifier-granting Doctrines, although the model does not assume that every extra draft contains one.

These are averages across complete simulated runs. Averages do not describe how widely individual outcomes vary; threshold probabilities and run-to-run variation are examined later in the report.

3.5. The role of Amplifiers in accelerating the chain

Amplifiers increase the rarity weight of the targeted Mission Control Blueprint. One targeting Amplifier gives a 2x weight multiplier; two give a 3x multiplier. The modifiers add, rather than multiply with one another.

Two acquisition conditions are now compared consistently across all five scenarios:

NATURAL: Both Amplifier-granting Doctrines must be acquired through the modeled reward system.

TIER I DOCTRINE FROM WAVE 1: The Tier I Amplifier-granting Doctrine (D1_45) is already owned at the start. It is removed from the later Doctrine pool; the Tier II Doctrine (D2_42) remains obtainable normally. This does not introduce a third Amplifier.

Average Mc Counts At Wave 60 (1,000 Runs)

Scenario	Natural	Tier I from W1
Baseline	7.147	7.574
Lions	16.187	17.903
Eleonor	11.338	12.336
Tander	10.007	10.811
Magel	10.290	10.472

Average Mc Counts At Wave 63 (1,000 Runs)

Scenario	Natural	Tier I from W1
Baseline	8.310	8.653
Lions	18.507	20.323
Eleonor	12.643	13.854
Tander	11.410	12.193
Magel	12.093	12.315

Granting the Tier I Doctrine from the beginning improves average acquisition for every scenario in this sample, but its impact is not uniform. The gain is particularly modest for Magel, whose extra Doctrine drafts already create more opportunities to find Amplifier-granting Doctrines through normal play.

The earlier 'saved Amplifier' comparison, which added a permanent artificial +100% bonus on top of naturally acquired Amplifiers, is superseded. Its previously reported Wave-60 value of 19.32 must not be reused for this revised scenario.

The separate no-Amplifier-effect ablation remains an earlier 500-run sensitivity test (Baseline Wave-60 mean 3.182). It is not directly part of the new matched 1,000-run scenario comparison and should be identified as such if retained.

SECTION 3 SUMMARY

Wave 40 marks the modeled shift from primarily reinforcement-driven growth to recursive Blueprint acquisition through Mission Control's 3A upgrade. Early Mission Controls and Amplifiers influence the opportunities available at that transition. By Wave 63, all four faction and commander bonuses outperform Baseline in average Mission Control ownership, with Lions leading the modeled scenarios. These results concern acquisition only; they do not establish a survival probability at the Wave 63–64 difficulty increase.

Model Notes

- All simulated MCs are immediately built; construction costs, placement limits, and held MC inventory are not modeled.

- Magel receives a choose-one-of-three Doctrine draft on even-numbered waves after ordinary reinforcement. Exact same-wave ordering remains provisional.

- The model prioritizes an Amplifier-granting Doctrine in Magel's draft when one is offered.

- The model's ordinary reward-selection policy does not systematically seek a second Amplifier after the first has been obtained.

- These numbers are from the 1,000-seed batch, not extrapolations.

SECTION 4 · FACTION AND COMMANDER COMPARISON

4.1. Baseline scenario

The baseline provides a reference for evaluating the benefits of faction- or commander-specific acquisition bonuses. It uses ordinary reinforcement rewards, standard Blueprint rarity progression, and the effects of acquired Mission Controls and Amplifiers.

Without additional scenario rewards, the strategy depends heavily on obtaining its first few Mission Controls through the normal acquisition process.

By Wave 40, the baseline averages 2.21 Mission Controls. Once 3A upgrades become available, the chain begins generating additional Blueprint opportunities, increasing the average to 7.15 by Wave 60 and 8.31 by Wave 63.

The baseline demonstrates that recursive acquisition can develop without a faction or commander bonus, although it generally progresses more slowly than the other scenarios.

4.2. Lions faction — Additional Blueprint acquisition

The Lions faction provides additional Blueprint quantities through ordinary reinforcement rewards.

This advantage directly supports Mission Control acquisition because a successful Blueprint reward can provide more than one copy.

Unlike bonuses that primarily improve the probability of finding a particular reward, Lions can increase the number of Mission Controls obtained when a qualifying reward is selected.

The effect is already evident by Wave 40, when Lions averages 4.81 Mission Controls, compared with 2.21 for the baseline.

After the 3A transition, Lions maintains its lead, reaching an average of 16.19 Mission Controls at Wave 60 and 18.51 at Wave 63.

Among the scenarios tested, Lions produces the highest average Mission Control count at every reported checkpoint.

4.3. Eleonor — Additional Blueprint opportunities

Eleonor receives an additional Blueprint draft every other wave, providing more opportunities to find Mission Control outside ordinary reinforcement.

These extra drafts improve the frequency with which the player can encounter desirable Blueprints.

Unlike Lions, Eleonor does not increase the quantity received from an ordinary Blueprint reward. Her advantage comes from having additional opportunities to select a Blueprint.

Eleonor averages 3.48 Mission Controls at Wave 40, increasing to 11.34 at Wave 60 and 12.64 at Wave 63.

Her performance places her second among the modeled scenarios at the main Wave 63 checkpoint.

The additional Blueprint opportunities remain valuable after Wave 40 because newly acquired Mission Controls can contribute their own 3A drafts.

4.4. Tander — Additional reinforcement choices

Tander increases the number of ordinary reinforcement cards from three to four.

This provides a broader selection of rewards at each reinforcement opportunity, improving the chance that a desirable card will be available.

The advantage is not limited to Mission Control itself. Additional choices can also help the player acquire supporting rewards, including Amplifier-granting Doctrines.

Tander averages 3.26 Mission Controls at Wave 40, 10.01 at Wave 60, and 11.41 at Wave 63.

This places Tander ahead of the baseline throughout the simulation, although behind Lions and Eleonor at the main checkpoint.

Tander's advantage is primarily one of selection flexibility: the player sees more alternatives without necessarily receiving more rewards from the chosen card.

4.5. Magel — Additional Doctrine opportunities

Magel receives an additional choose-one-of-three Doctrine draft every other wave.

Unlike Eleonor's Blueprint drafts, these rewards do not directly offer Mission Control. Instead, Magel gains additional opportunities to acquire Doctrines that can improve the acquisition process.

For this analysis, the most relevant rewards are the two Doctrines that grant Amplifiers.

Obtaining an Amplifier earlier increases the number of subsequent Blueprint opportunities that can benefit from its weight bonus.

This gives Magel a more indirect acquisition advantage than the other scenarios.

At Wave 40, Magel averages 2.82 Mission Controls, placing him behind Lions, Eleonor, and Tander.

However, his relative performance improves as the run progresses.

By Wave 60, Magel averages 10.29 Mission Controls, slightly exceeding Tander's 10.01. At Wave 63, Magel reaches 12.09, compared with Tander's 11.41.

Magel therefore demonstrates that additional Doctrine opportunities can support Mission Control growth even without directly increasing the number of Blueprint drafts.

His modeled performance remains subject to uncertainty about the exact ordering and selection behavior of his additional Doctrine rewards.

4.6. Comparative results

Average Mission Controls Acquired

Natural Amplifier acquisition; 1,000 simulations per scenario.

Scenario	Wave 40	Wave 60	Wave 63	Wave 83	Wave 103
Baseline	2.21	7.15	8.31	16.69	27.31
Lions	4.81	16.19	18.51	36.45	56.56
Eleonor	3.48	11.34	12.64	26.01	40.87
Tander	3.26	10.01	11.41	22.92	35.76
Magel	2.82	10.29	12.09	27.43	46.17

The results show two important patterns.

First, the faction and commander bonuses produce differences well before Wave 40. These early advantages affect the number of Mission Controls available when 3A upgrades become accessible.

Second, the relative performance of scenarios can change as the run progresses. Magel is initially behind Tander, but overtakes him before Wave 60 and extends that advantage during the later stages.

At Wave 83, Magel averages 27.43 Mission Controls compared with Eleonor's 26.01. At Wave 103, Magel averages 46.17 compared with Eleonor's 40.87 and Tander's 35.76.

These are average acquisition counts, not measures of combat effectiveness. A scenario producing more Mission Controls does not necessarily provide a higher probability of surviving a particular wave.

SECTION 4 SUMMARY

Lions provides the strongest Mission Control acquisition performance in the current model, benefiting from additional Blueprint quantities in ordinary reinforcement.

Eleonor performs particularly well before Wave 63 through additional Blueprint opportunities.

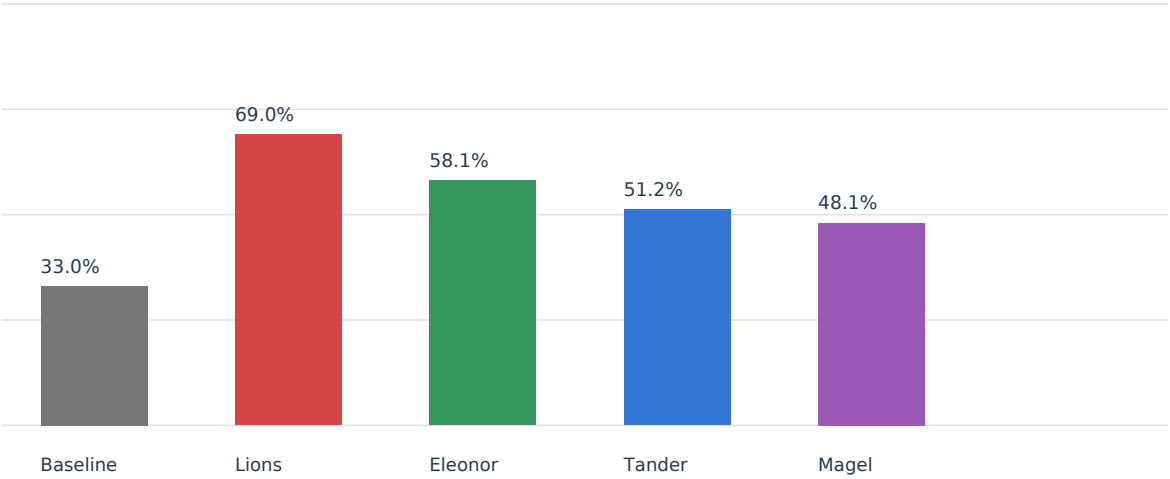
Tander improves the selection of ordinary reinforcement rewards, providing a consistent advantage over the baseline.

Magel follows a different pattern. His additional Doctrine drafts support Amplifier acquisition, allowing his relative performance to improve over time. Although he starts behind Tander, he overtakes him by Wave 60 and exceeds Eleonor's average by Wave 83.

The next section examines these differences at Wave 63, focusing not only on average Mission Control counts but also on the probability of reaching specific acquisition thresholds.

SECTION 5 · THE WAVE 63 SURVIVAL THRESHOLD

FIGURE 1 · Wave 63: probability of acquiring at least 10 MCs



Natural acquisition; 1,000 runs per scenario. This is not a survival probability.

5.1. Why Wave 63 matters

Wave 63 is the principal evaluation point because enemy wave size increases substantially around Waves 63–64.

A Mission Control strategy is most useful when its additional defensive options become available before that increase. Wave 40 introduces 3A Blueprint drafts, leaving the period from Waves 40 to 63 for the chain to develop.

A high Mission Control count reached much later demonstrates long-term acquisition potential, but does not necessarily help the player overcome the earlier transition.

5.2. Mission Control acquisition before Wave 63

Average Mc Count — Natural Amplifier Acquisition

Scenario	W40	W50	W60	W63
Baseline	2.21	4.24	7.15	8.31
Lions	4.81	9.91	16.19	18.51
Eleonor	3.48	6.76	11.34	12.64
Tander	3.26	6.02	10.01	11.41
Magel	2.82	5.74	10.29	12.09

All five scenarios show substantial growth between Waves 40 and 63. Lions maintains the highest average throughout. Eleonor remains second at Wave 63, while Magel moves ahead of Tander between Waves 50 and 60.

Averages alone do not show whether most runs reach a similar count or whether high-count runs pull the average upward. For this reason, the analysis also compares acquisition thresholds.

5.3. Probability of reaching Mission Control thresholds

The thresholds of 5, 10, and 20 Mission Controls are analytical reference points, not verified survival requirements.

Each probability is the share of the 1,000 simulated runs that acquired at least the indicated count by the end of Wave 63.

Wave 63 Threshold Probabilities — Natural Acquisition

Scenario	>=5 MC	>=10 MC	>=20 MC
Baseline	67.0%	33.0%	6.1%
Lions	86.4%	69.0%	36.5%
Eleonor	84.9%	58.1%	16.7%
Tander	82.9%	51.2%	13.7%
Magel	76.5%	48.1%	18.9%

Lions leads at every threshold, especially at 20 Mission Controls, reached in 36.5% of runs.

Eleonor reaches at least ten in 58.1% of runs, followed by Tander at 51.2%.

Magel has a higher average than Tander but a lower chance of reaching five or ten. Conversely, Magel reaches twenty in 18.9% of runs, compared with 13.7% for Tander and 16.7% for Eleonor.

This indicates that Magel's higher average is associated with more high-count outcomes, not a uniformly higher likelihood of reaching every threshold.

5.4. Scenario comparison at Wave 63

Two different qualities matter: how frequently a strategy establishes a moderate chain, and how often it produces a particularly large chain.

Lions leads on both measures. Eleonor and Tander perform strongly at moderate thresholds. Magel is less consistent at reaching five or ten, but more frequently produces runs with twenty or more Mission Controls than either Eleonor or Tander.

Magel's additional Doctrine opportunities are an indirect acquisition advantage. Their value depends on which Doctrines are offered and when Amplifiers become available.

The baseline trails the scenarios, particularly at ten and twenty Mission Controls.

5.5. The effect of beginning with the Tier I Amplifier Doctrine

The alternative scenario starts with the Tier I Amplifier Doctrine already owned at Wave 1. That Doctrine is removed from later reward pools, while the Tier II Doctrine remains naturally obtainable. The maximum remains two Amplifiers.

Wave 63: Probability Of >=10 Mc

Scenario	Natural	Tier I from W1	Difference
Baseline	33.0%	36.1%	+3.1 pp
Lions	69.0%	73.7%	+4.7 pp
Eleonor	58.1%	65.0%	+6.9 pp
Tander	51.2%	56.7%	+5.5 pp
Magel	48.1%	50.4%	+2.3 pp

Every scenario improves in this sample. Eleonor has the largest gain in this measure, and Magel the smallest.

The modest change for Magel is consistent with his extra Doctrine drafts already providing additional opportunities to obtain Amplifier-granting Doctrines naturally. This is a plausible interpretation of the model results, not a separately isolated causal test.

5.6. Interpreting the results in relation to survival

These acquisition probabilities must not be presented as survival probabilities.

The model does not calculate enemy damage, defensive effectiveness, resource availability, or whether every acquired building can be constructed and upgraded in time. Nor does it establish a minimum MC count needed to survive Waves 63–64.

A run with fewer Mission Controls may still have an effective defense. A run with many may lack the necessary resources or supporting buildings.

The simulation instead estimates how frequently each scenario reaches different acquisition levels before the difficulty transition under its stated assumptions.

SECTION 5 SUMMARY

Wave 63 is the practical pre-escalation checkpoint. Lions reaches at least ten MCs in 69.0% of natural-acquisition runs and at least twenty in 36.5%.

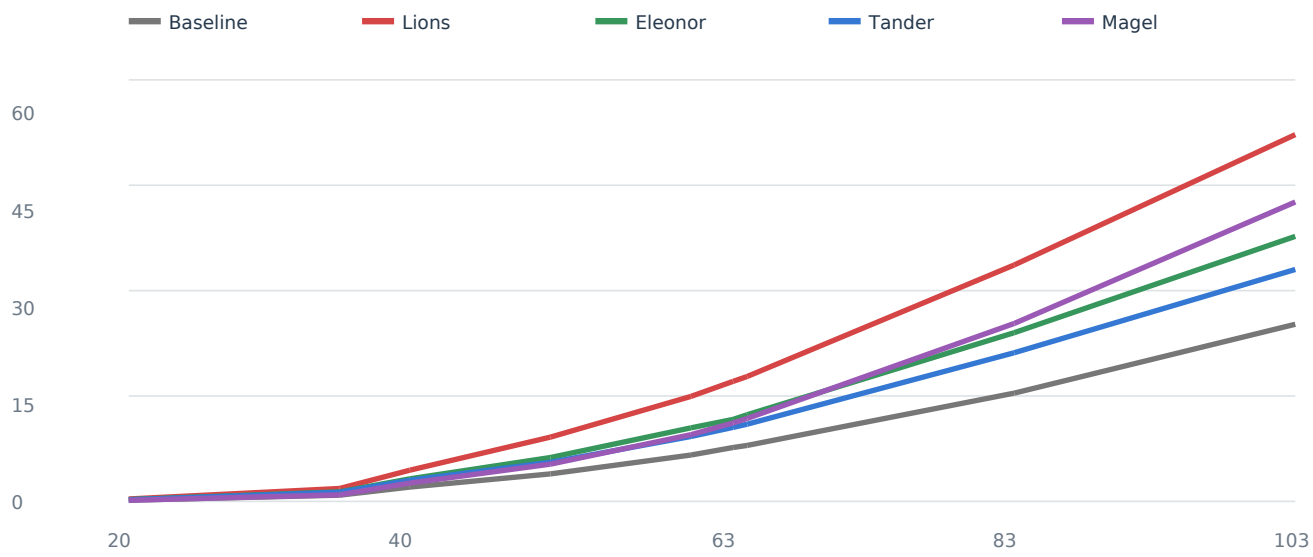
Eleonor and Tander have strong moderate-threshold performance. Magel has more high-count outcomes, even though it reaches five or ten less often than those scenarios.

Guaranteeing the Tier I Amplifier Doctrine improves the ten-MC threshold for every commander, with the smallest gain for Magel.

Threshold probabilities reveal differences that averages alone conceal. They remain measures of acquisition, not survival.

SECTION 6 · LATE-GAME GROWTH: WAVES 64–103

FIGURE 2 · Mean Mission Controls by recorded checkpoint



Natural acquisition. Lines connect recorded checkpoints only; no intermediate results are implied.

6.1. Growth after Wave 63

The simulation continues through Wave 103 under the same acquisition assumptions. It does not model combat survival, so these results describe growth potential in runs allowed to continue rather than expected outcomes among actual surviving players.

As additional Mission Controls are acquired, their 3A Blueprint drafts create further opportunities to extend the chain.

Average Mission Controls — Natural Amplifier Acquisition

Scenario	W63	W64	W83	W103
Baseline	8.31	8.63	16.69	27.31
Lions	18.51	19.26	36.45	56.56
Eleonor	12.64	13.36	26.01	40.87
Tander	11.41	11.90	22.92	35.76
Magel	12.09	12.79	27.43	46.17

Lions leads throughout the late game. Magel shows strong growth and moves ahead of Eleonor in average Mission Control count by Wave 83. Baseline continues growing but remains behind the scenarios.

6.2. Comparing late-game growth rates

The change in average MC count between Waves 63 and 103 is calculated directly from the simulated checkpoint averages, not from an extrapolated growth model.

Scenario	W63	W103	Change
Baseline	8.31	27.31	+19.00
Lions	18.51	56.56	+38.05
Eleonor	12.64	40.87	+28.22
Tander	11.41	35.76	+24.35
Magel	12.09	46.17	+34.07

Lions adds the most Mission Controls over the interval. Magel has the second-largest increase, approximately 34.07, exceeding Eleonor's 28.22 despite starting Wave 63 slightly behind.

This is consistent with Magel's additional Doctrine opportunities becoming influential over longer runs. The comparison does not isolate the contribution of each individual Doctrine.

6.3. Changes in scenario rankings

Average MC count at Wave 63:

1. Lions
2. Eleonor
3. Magel
4. Tander
5. Baseline

Average MC count at Waves 83 and 103:

1. Lions
2. Magel
3. Eleonor
4. Tander
5. Baseline

Magel's move ahead of Eleonor is the principal ranking change. Scenario comparisons therefore depend on the checkpoint: Eleonor performs better at Wave 63, while Magel has the larger average by Wave 83.

6.4. Late-game acquisition thresholds

Probability Of >=20 Mc — Natural Amplifier Acquisition

Scenario	W63	W83	W103
Baseline	6.1%	29.7%	65.3%
Lions	36.5%	78.5%	95.6%
Eleonor	16.7%	63.2%	91.7%
Tander	13.7%	52.7%	86.7%
Magel	18.9%	59.5%	91.4%

Although Magel has a higher mean than Eleonor at Waves 83 and 103, Eleonor retains a slightly higher probability of reaching the 20-MC threshold at those checkpoints.

This illustrates that larger average chains do not imply greater reliability at every threshold. Lions remains strongest on this measure.

6.5. Late-game results with the Tier I Amplifier Doctrine

The alternative condition grants the Tier I Amplifier Doctrine at Wave 1, removes it from later rewards, and leaves the Tier II Doctrine available naturally.

Average Mc Count At Wave 103

Scenario	Natural	Tier I from W1	Change
Baseline	27.312	28.583	+1.271
Lions	56.556	60.584	+4.028
Eleonor	40.866	43.008	+2.142
Tander	35.762	36.828	+1.066
Magel	46.167	46.586	+0.419

All five scenarios have higher average counts under the saved-Doctrine condition. Lions has the largest absolute gain, and Magel the smallest.

The small change for Magel does not mean Amplifiers are unimportant. It indicates that granting this specific Doctrine at the start provides relatively little incremental benefit compared with Magel's natural modeled acquisition opportunities.

6.6. Practical interpretation and limitations

All simulated runs continue through Wave 103, regardless of whether their defenses could actually survive enemy attacks.

The results therefore measure acquisition potential, not expected MC counts among real survivors. This matters particularly for Magel, whose advantages become more pronounced late in the run.

Similarly, Lions' leading acquisition results do not establish that Lions has the strongest overall combat strategy.

SECTION 6 SUMMARY

Lions maintains the highest average, reaching 56.56 MC by Wave 103 under natural acquisition.

Magel has the second-largest post-Wave-63 increase and overtakes Eleonor in average count by Wave 83, reaching 46.17 by Wave 103.

Eleonor nevertheless remains slightly more consistent at the 20-MC threshold at Waves 83 and 103.

The Tier I-from-Wave-1 scenario improves averages for every commander, with the smallest incremental gain for Magel.

These results describe long-term acquisition characteristics; Wave 63 remains the main strategic checkpoint.

SECTION 7 · AMPLIFIER ANALYSIS

7.1. How Amplifiers affect Blueprint acquisition

Each Amplifier assigned to Mission Control adds +100% to its Blueprint rarity weight. Bonuses stack additively:

0 Amplifiers: 1x weight

1 Amplifier: 2x weight

2 Amplifiers: 3x weight

Rarity weight is not the same as acquisition probability. Mission Control's probability depends on the combined weights of eligible Blueprints, including competing Tier 2 and Tier 3 Blueprints affected by Mission Control's 1A upgrade. The effect therefore depends on the current reward pool.

7.2. Natural Amplifier acquisition

Two independent Doctrines grant Amplifiers:

Tier I D1_45: one Amplifier.

Tier II D2_42: another Amplifier.

Both can be acquired naturally, but neither is guaranteed. Early acquisition allows the bonus to affect more later Blueprint opportunities.

An Amplifier acquired before the first Mission Control becomes useful when a Mission Control is available to receive its effect.

The current reward-selection policy prioritizes an initial Amplifier but does not systematically seek a second after obtaining the first. This is a model limitation, not a game mechanic.

7.3. Guaranteed Tier I Doctrine from Wave 1

The alternative condition begins with D1_45 already owned at Wave 1, removes it from later Doctrine rewards, and leaves D2_42 naturally obtainable.

The maximum remains two Amplifiers, for a 3x weight multiplier.

This supersedes the earlier artificial saved-Amplifier scenario, which applied a permanent additional weight bonus on top of naturally acquired Amplifiers.

7.4. Effect on average Mission Control acquisition

Average Mc Count At Wave 63 — 1,000 Runs Per Condition

Scenario	Natural	Tier I from W1	Change
Baseline	8.310	8.653	+0.343
Lions	18.507	20.323	+1.816
Eleonor	12.643	13.854	+1.211
Tander	11.410	12.193	+0.783
Magel	12.093	12.315	+0.222

Every scenario has a higher average with the Tier I Doctrine guaranteed. Lions has the largest absolute increase at Wave 63, and Magel the smallest.

7.5. Effect on acquisition reliability

Wave 63 — Probability Of >=10 Mc

Scenario	Natural	Tier I from W1	Change
Baseline	33.0%	36.1%	+3.1 pp
Lions	69.0%	73.7%	+4.7 pp
Eleonor	58.1%	65.0%	+6.9 pp
Tander	51.2%	56.7%	+5.5 pp
Magel	48.1%	50.4%	+2.3 pp

Eleonor has the largest improvement at this threshold, while Magel has the smallest.

Wave 63 — Probability Of >=20 Mc

Scenario	Natural	Tier I from W1	Change
Baseline	6.1%	6.1%	+0.0 pp
Lions	36.5%	43.6%	+7.1 pp
Eleonor	16.7%	20.6%	+3.9 pp
Tander	13.7%	15.4%	+1.7 pp
Magel	18.9%	19.4%	+0.5 pp

Lions has the largest improvement at the 20-MC threshold. Baseline shows no difference at this threshold in the 1,000-run sample, despite improvements in its mean and 10-MC threshold.

7.6. Why Magel responds differently

Magel's additional choose-one-of-three Doctrine draft every other wave creates more opportunities to encounter Amplifier-granting Doctrines naturally.

Average Mc Count — Magel

Checkpoint	Natural	Tier I from W1	Change
Wave 40	2.822	2.959	+0.137
Wave 60	10.290	10.472	+0.182
Wave 63	12.093	12.315	+0.222
Wave 103	46.167	46.586	+0.419

These modest differences are consistent with the hypothesis that Magel already has relatively good natural access to Amplifiers.

However, the current results do not separately measure the probability or timing of each Doctrine's acquisition. This is an interpretation, not an isolated causal measurement.

7.7. Amplifier sensitivity and model limitations

An earlier 500-run sensitivity experiment gave a Baseline Wave-60 mean of 3.182 MCs with Amplifier effects disabled, compared with 7.172 in the earlier natural-acquisition batch.

These are prior-batch figures, not directly matched 1,000-run results, and must not be conflated with the current comparison.

A future controlled analysis could compare zero, one, and two Amplifiers under the same 1,000 seeds and current model implementation, recording when each Amplifier is obtained and applied.

Until then, distinguish the measured effect of guaranteeing Tier I access from the broader causal effect of Amplifiers.

SECTION 7 SUMMARY

Amplifiers increase Mission Control's Blueprint rarity weight and can accelerate chain growth.

Guaranteeing the Tier I Doctrine from Wave 1 improves average acquisition in all five current scenarios. Lions gains the most MCs on average at Wave 63, Eleonor gains the most at the 10-MC threshold, and Magel has the smallest incremental gain.

These results measure guaranteed early Tier I access, not the isolated effect of each individual Amplifier or its acquisition timing.

SECTION 8 · CONCLUSIONS

The simulations demonstrate substantial Blueprint acquisition growth from Mission Control chains, particularly after the 3A upgrade becomes available at Wave 40. Timing and reliability of acquisition matter as much as the eventual chain size.

8.1. Main findings

Lions — Strongest Overall Acquisition

Lions leads in average Mission Control count at every evaluated checkpoint. At Wave 63 it averages 18.51 MCs versus 8.31 for Baseline, and has the highest probabilities of reaching at least 10 or 20 MCs. Its additional Blueprint quantities create a direct acquisition advantage.

Eleonor — Strong Early Reliability

Eleonor reaches at least ten MCs by Wave 63 in 58.1% of natural-acquisition runs, compared with 51.2% for Tander and 48.1% for Magel. Her extra Blueprint drafts help establish the chain without depending on specific supporting Doctrines.

Tander — Consistent Reinforcement Advantage

Tander's fourth reinforcement card improves reward selection. This yields higher mean counts and threshold probabilities than Baseline throughout the simulation, though it does not lead the principal scenario comparisons.

Magel — Increasing Late-Game Strength

Magel's additional Doctrine drafts produce an indirect acquisition advantage. He begins behind the other scenarios at Wave 40, overtakes Tander in mean count by Wave 60, and overtakes Eleonor by Wave 83. At Wave 103 Magel averages 46.17 MCs versus Eleonor's 40.87. However, Eleonor retains a slightly higher probability of reaching 20 MCs at Wave 103.

8.2. Wave 63 remains the primary strategic checkpoint

Relative scenario advantages depend on the evaluation wave. High late-game counts may not help overcome the Wave 63–64 increase in enemy wave size.

Wave 63 threshold probabilities are therefore more relevant to the report's main question than maximum late-game chain sizes.

The simulation does not establish the number of MCs required to survive Wave 63. It measures how frequently different acquisition levels are reached by then.

8.3. Amplifiers contribute to chain development

Amplifiers increase Mission Control Blueprint rarity weight. Guaranteeing the Tier I Amplifier Doctrine from Wave 1 raises average MC counts for all five scenarios.

Lions has the largest absolute gain in mean MC count at Wave 63. Eleonor has the largest gain in the probability of reaching at least ten. Magel has the smallest incremental gain, consistent with his additional natural Doctrine opportunities.

The comparison does not independently isolate each Amplifier's contribution or acquisition timing.

8.4. Implications for faction and commander selection

Highest average MC count at Wave 63: Lions.

Highest probability of >=10 MCs at Wave 63: Lions.

Highest probability of >=20 MCs at Wave 63: Lions.

Strongest non-Lions result at >=10 MCs by Wave 63: Eleonor.

Highest late-game mean among non-Lions scenarios: Magel.

Smallest incremental gain from guaranteed Tier I Doctrine: Magel.

These rankings concern acquisition only, not combat effectiveness, actual survival, or overall scenario strength.

8.5. Limitations and future work

Important simplifications include immediate MC construction, the assumption that 3A upgrades are affordable beginning at Wave 40, and omission of combat survival and complete resource management.

Pit, Reactive, and Calculator effects are not included.

Future work could test Amplifier acquisition timing, alternative reward-selection policies, resource constraints, and sensitivity to the Wave 40 upgrade assumption. Simulation parameters should not be tuned to force agreement with observed gameplay.

8.6. Final conclusion

Lions is the strongest Mission Control acquisition scenario in the current simulation. Eleonor offers strong pre-Wave-63 acquisition reliability, while Magel becomes more competitive over longer runs. Tander consistently improves on Baseline through additional reinforcement choices.

The principal strategic finding is that establishing the chain early enough matters more than achieving a high MC count late in the run.

These simulations quantify acquisition potential under specified assumptions; they do not establish a direct relationship between MC count and survival.

APPENDIX · SIMULATION PARAMETERS AND REPRODUCIBILITY

A.1. Dataset

File: SIMULATION-1000-RUNS.json

Scenarios: Baseline, Lions, Eleonor, Tander, Magel.

Conditions: Natural Doctrine acquisition; Tier I Amplifier Doctrine preowned from Wave 1.

Runs per scenario: 1,000, seeds 1–1000.

Total runs: 10,000.

Simulated waves per run: 103.

Recorded checkpoints: Waves 20, 35, 40, 50, 60, 63, 64, 83, 103.

Recorded statistics: mean MC count; percentage with >=5, >=10, >=20 MCs.

No censored runs were accepted by the batch runner.

DATA COMPATIBILITY: The raw JSON retains the historical scenario key 'Eleanor', corresponding to the display name 'Eleonor'. This is a label alias only; numeric data is unchanged. Report generation should normalize the display name, not alter results.

A.2. Model implementation

Repository: /home/pcastaneda/Documents/tower-dominion-analysis

Simulation implementation: model/mc_wave60.py

Blueprint roster: model/mc_roster.py, maps/blueprints.json

Doctrine and draft logic: model/mc_doctrines.py and supporting model modules.

Batch runner: tools/mc_1000_run.py (copied from the completed batch runner).

The runner uses the same seeds for each scenario and records all selected checkpoints. The result JSON was captured from the completed batch.

A.3. Scenario configurations

Baseline: no faction- or commander-specific acquisition modifier.

Lions (faction): additional Blueprint quantities from ordinary reinforcement.

Eleonor (commander): periodic additional Blueprint draft.

Tander (commander): four reinforcement cards instead of three.

Magel (commander): additional choose-one-of-three Doctrine draft every other wave.

Magel's extra draft occurs after ordinary reinforcement in the current implementation; exact same-wave ordering is provisional.

A.4. Core acquisition rules

32 ordinary reinforcement-eligible Blueprints: 12 Tier I, 10 Tier II, 10 Tier III.

Ordinary reinforcement: three cards of two rewards, with Tander receiving four cards.

MC 1A: +2 raw rarity weight to eligible Tier II and Tier III Blueprints for each MC.

MC 3A: two choose-one-of-five Blueprint drafts per MC.

Wave 40 onward: model assumes every constructed MC can be upgraded to 3A.

Newly acquired MCs are built and their available drafts resolved immediately.

One assigned Amplifier: 2x MC rarity weight. Two: 3x (additive +100% each).

Natural condition: D1_45 and D2_42 can be acquired from rewards.

Saved Tier I condition: D1_45 is preowned at Wave 1 and removed from reward pool; D2_42 remains obtainable.

A.5. Scope and exclusions

The model does not simulate combat, survival, building placement, actual resource budgets, or holding acquired MCs unbuilt. Pit, Reactive, and Calculator effects are excluded. Some reward sequencing and scenario details remain provisional.

The simulation's reward policy does not systematically prioritize the second Amplifier after the first.

The thresholds >=5, >=10, and >=20 MCs are analytical markers, not established combat survival requirements.

A.6. Statistical interpretation

Each reported percentage is an empirical share of the 1,000 simulated runs. A difference of 1 percentage point corresponds to 10 runs. No confidence intervals or formal significance tests have been calculated.

The Wave 63 and Wave 103 averages are unconditional on actual survival, because combat is not modeled.

The earlier 500-run no-Amplifier sensitivity result is a separate batch and should not be treated as matched to the current 1,000-run scenarios.

A.7. Reproduction

From the repository root:

PYTHONPATH=model python3 tools/mc_1000_run.py

The runner writes its result file to /tmp/td_mc_1000_results.json and prints summary lines to standard output. Reproducing the recorded JSON requires the same model and input roster versions; repository changes may affect results. The simulation source and runner are presently uncommitted.

A.8. Presentation workflow

The authoritative editorial inputs are REPORT-SECTION-01.txt through REPORT-SECTION-08.txt and this Appendix.

The authoritative numeric input is SIMULATION-1000-RUNS.json.

Generate charts and final PDF/HTML only after the text has been reviewed; never reverse-engineer the source from rendered presentation files.